

# 1 Pre-training of diffusion model

Forward SDE (data to noise):

$$dX_t = f(X_t, t)dt + g(t)dW_t, \quad X_0 \sim p_{data}.$$

In practice,  $f, g$  are usually chosen as below:

$$f(x, t) = -\frac{1}{2}\beta(t)x, \quad g(t) = \sqrt{\beta(t)}.$$

The training objective is

$$\begin{aligned} \mathcal{J}(\theta) &= \mathbb{E}_{t \sim U(0, T)} \mathbb{E}_{x \sim p_t} [\omega(t) \|s_\theta(x, t) - \nabla_x \log p_t(x)\|_2^2] \\ &\equiv \mathbb{E}_t \mathbb{E}_{x_0 \sim p_0, x_t \sim p_t(\cdot|x_0)} [\omega(t) \|s_\theta(x_t, t) - \nabla_{x_t} \log p_t(x_t, x_0)\|_2^2]. \end{aligned}$$

To compute this loss function and take gradient w.r.t.  $\theta$ , we need  $x_t, t$  pairs and the conditional score  $\nabla_{x_t} \log p_t(x_t|x_0)$ . With the choice of  $f, g$  above, under linear drift, the  $X_t|X_0$  is Gaussian

$$X_t = \alpha(t)X_0 + \sigma(t)\varepsilon, \quad \alpha(t) = \exp(-\frac{1}{2} \int_0^t \beta), \sigma(t) = \sqrt{1 - \alpha(t)^2}.$$

Then we could compute the conditional score

$$\nabla_{x_t} \log p_t(x_t|x_0) = -\frac{x_t - \alpha(t)x_0}{\sigma(t)^2}.$$

The training weight at each time step  $\omega(t)$  is usually chosen to be  $\sigma^2$  or  $\beta$ . A popular choice of  $\beta$  (used in generation phase) and corresponding  $\alpha$  (used in training phase) is the cosine schedule given in (Nichol & Dhariwal, 2021):

$$\begin{aligned} t = 0, 1, \dots, T, \quad f(t) &= \cos^2 \left( \frac{\pi}{2} \frac{t/T + s}{1 + s} \right), \quad s = 0.008 \ll 1, \\ \bar{\alpha}_t &= f(t)/f(0), \quad t = 0, \dots, T, \\ \alpha_t &= \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}, \quad \beta_t = 1 - \alpha_t, \quad t = 1, \dots, T. \end{aligned}$$

Since  $\nabla_x \log p_t(x)$  is unknown, we use the denoising score matching (DSM) equivalence. Backward SDE:

$$\begin{aligned} dX_t &= \left[ -f(X_t, t) + \frac{1 + \eta^2}{2} g(t)^2 s_\theta(X_t, t) \right] dt + \eta g(t) dW_t, \quad t : T \rightarrow 0. \\ x_{k+1} &= x_k + \left[ -f(x_k, t_k) + \frac{1 + \eta^2}{2} g(t_k)^2 s_\theta(x_k, t_k) \right] \Delta t + \eta g(t_k) \sqrt{-\Delta t} z_k, \quad z_k \sim \mathcal{N}(0, I_d), \end{aligned}$$

where  $\Delta t < 0, T = t_0 > t_1 > \dots > t_N = 0$ . Initialize  $x_T$  from a standard Gaussian. If  $\eta = 1$ , then we use stochastic reverse-time SDE (maximum diversity). If  $\eta = 0$ , we use probability flow ODE (deterministic, but often fastest if employing external ODE solvers).

## 2 Pretraining

### 2.1 Forward (training) process and schedule

#### 2.1.1 - Forward SDE (variance-preserving/VP):

$$dX_t = f(X_t, t) dt + g(t) dW_t, \quad f(x, t) = -\frac{1}{2}\beta(t)x, \quad g(t) = \sqrt{\beta(t)}.$$

The forward marginal (for data  $X_0$ ) is

$$X_t = \alpha(t)X_0 + \sigma(t)\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I),$$

with

$$\alpha(t) = \exp\left(-\frac{1}{2}\int_0^t \beta(u) du\right), \quad \sigma(t) = \sqrt{1 - \alpha(t)^2}, \quad \beta(t) = -\frac{d}{dt} \log \alpha(t)^2.$$

(Equivalently,  $\alpha(t)^2 = \exp\left(-\int_0^t \beta(u) du\right)$ .) (Song et al., 2021)

#### 2.1.2 - Cosine schedule (Nichol-Dhariwal):

choose

$$\bar{\alpha}(t) = \frac{f(t)}{f(0)}, \quad f(t) = \cos^2\left(\frac{\pi}{2} \frac{t+s}{1+s}\right), \quad s \approx 0.008$$

then  $\alpha(t) = \sqrt{\bar{\alpha}(t)}$ ,  $\sigma(t) = \sqrt{1 - \bar{\alpha}(t)}$ . (A continuous  $\beta(t)$  follows from  $\beta(t) = -\frac{d}{dt} \log \bar{\alpha}(t)$ .) (Nichol and Dhariwal, 2021)

### 2.2 Training objective (DSM / score matching on VP)

Sample  $t \sim \mathcal{U}[0, 1]$ ,  $\varepsilon \sim \mathcal{N}(0, I)$ , form  $x_t = \alpha(t)x_0 + \sigma(t)\varepsilon$ . Train a score network  $s_\theta(x, t) \approx \nabla_x \log p_t(x)$  with the denoising score-matching loss

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \varepsilon} \left[ \lambda(t) \|s_\theta(x_t, t) - s^*(x_t, t)\|_2^2 \right], \quad s^*(x_t, t) = \nabla_{x_t} \log p_t(x_t | x_0) = -\frac{x_t - \alpha(t)x_0}{\sigma(t)^2},$$

with a time weight  $\lambda(t)$  (common choices:  $\lambda(t) = \sigma(t)^2$  or  $\lambda(t) = \beta(t)$ ). (Song et al., 2021)

## 3 Sampling

### 3.1 Sampling method One- Reverse-time SDE (stochastic)

Given the learned score, the reverse-time SDE (integrated from  $t = 1 \rightarrow 0$ ) is

$$dX_t = [f(X_t, t) - g(t)^2 s_\theta(X_t, t)] dt + g(t) d\bar{W}_t,$$

which you can discretize on a decreasing grid  $t_0 = 1 > \dots > t_N = 0$  (Euler-Maruyama,  $\Delta t_k = t_{k+1} - t_k < 0$ ):

$$x_{k+1} = x_k + \left[ f(x_k, t_k) - g(t_k)^2 s_\theta(x_k, t_k) \right] \Delta t_k + g(t_k) \sqrt{-\Delta t_k} z_k, \quad z_k \sim \mathcal{N}(0, I).$$

For VP, substitute  $f(x, t) = -\frac{1}{2}\beta(t)x$ ,  $g(t)^2 = \beta(t)$ . Initialize  $x_{t_0} \sim \mathcal{N}(0, I)$ . (Song et al., 2021)

### 3.2 Sampling method two - Probability-flow ODE (deterministic)

The probability-flow ODE has the same time-marginals as the reverse SDE and reads

$$\frac{dX_t}{dt} = f(X_t, t) - \frac{1}{2}g(t)^2 s_\theta(X_t, t),$$

with an explicit-Euler step

$$x_{k+1} = x_k + \left[ f(x_k, t_k) - \frac{1}{2}g(t_k)^2 s_\theta(x_k, t_k) \right] \Delta t_k, \quad \Delta t_k < 0.$$

Again, plug  $f(x, t) = -\frac{1}{2}\beta(t)x$ ,  $g(t)^2 = \beta(t)$ ; start from  $x_{t_0} \sim \mathcal{N}(0, I)$ . (Song et al., 2021)

(Unification you’ve been using.) A convenient one-liner that covers both is

$$x_{k+1} = x_k + \left[ f - \frac{1 + \eta^2}{2} g^2 s\theta \right] \Delta t_k + \eta g \sqrt{-\Delta t_k} z_k, \quad \eta \in \{0, 1\},$$

with  $\eta = 1$  giving the reverse SDE and  $\eta = 0$  the PF-ODE. (Song et al., 2021)

Faster deterministic sampling with the PF-ODE, you can swap Euler for DPM-Solver / DPM-Solver++. (Lu et al., 2022)

### 3.3 Prefix Constraint (Mask Projection)

Let the first  $K$  coordinates be ”known” (your fixed prefix  $v \in \mathbb{R}^K$ ); define a mask  $M = \text{diag}(\underbrace{1, \dots, 1}_K, \underbrace{0, \dots, 0}_{D-K})$ . After each step, project:

$$x_{k+1} \leftarrow M X_{\text{fix}}(t_{k+1}) + (I - M)x_{k+1},$$

with - SDE version:  $X_{\text{fix},1:K}(t) = \alpha(t)v + \sigma(t)\varepsilon_t, \varepsilon_t \sim \mathcal{N}(0, I_K)$ ; - PF-ODE version:  $X_{\text{fix},1:K}(t) = \alpha(t)v$ .

Since  $\sigma(0) = 0$ , you end at  $x_{0,1:K} = v$  exactly. (This is the standard inpainting/conditioning trick used in DDPM/score-SDE works such as RePaint.) (Lugmayr et al., 2022)

## References

- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. DPM-solver: A fast ODE solver for diffusion probabilistic model sampling in around 10 steps. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, 2022. URL [https://openreview.net/forum?id=2uAaGw1P\\_V](https://openreview.net/forum?id=2uAaGw1P_V).
- Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. Repaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11461–11471, 2022.
- Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8162–8171. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/nichol21a.html>.
- Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=PxtIG12RRHS>.